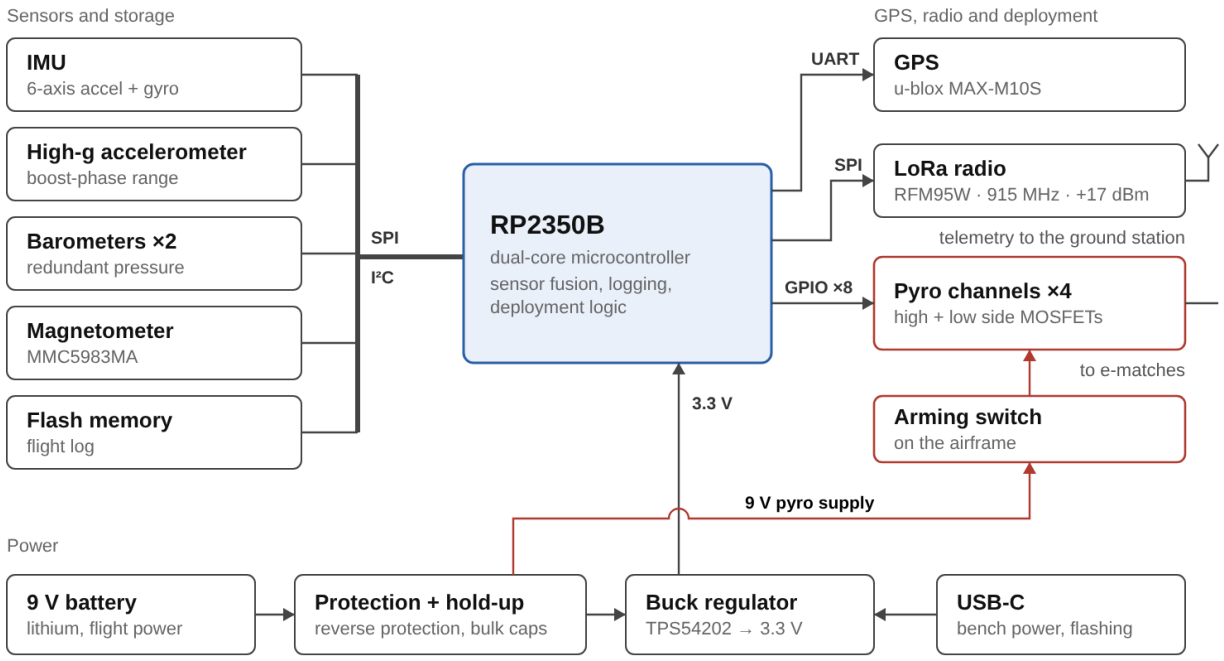
DYLAN PIWKO

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KESTREL FLIGHT COMPUTER

Designer, Personal Project
October 2026 – Present



Kestrel block diagram

What?

- Flight computer for my 4" Wildman Punisher 4, which I'm flying for my NAR Level 2 certification in January 2027
- It logs data and sends live telemetry, and will eventually handle dual-deployment recovery, including a later L2200G flight that simulates to about 14,800 ft and Mach 1.7

How?

- Designed the schematic and 4-layer board in **KiCad** (100 × 40 mm, 145 parts)
- RP2350B microcontroller, IMU, high-g accelerometer, two barometers, magnetometer, u-blox GPS, 915 MHz LoRa radio and flash memory
- Four pyro channels switched on both the high and low side, behind a physical arming switch
- 9 V input with reverse protection and a TPS54202 buck converter for 3.3 V
- Wrote bring-up firmware and a desktop ground station with a live map

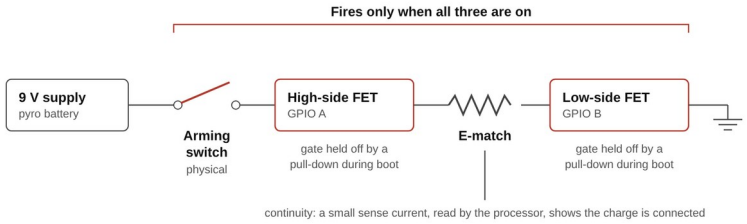
Results

- Zero ERC/DRC errors, with every part checked pin by pin against its datasheet and Raspberry Pi's reference design
- Fixed everything an outside schematic review flagged
- PCBWay is building the boards through their Future Innovators sponsorship
- My commercial altimeters keep handling recovery until Kestrel passes ground and flight tests

DESIGN AND VERIFICATION

Each pyro channel switches the e-match on both its high side and its low side, behind a physical arming switch on the airframe. A charge can only fire when the switch is closed and the processor turns on both transistors.

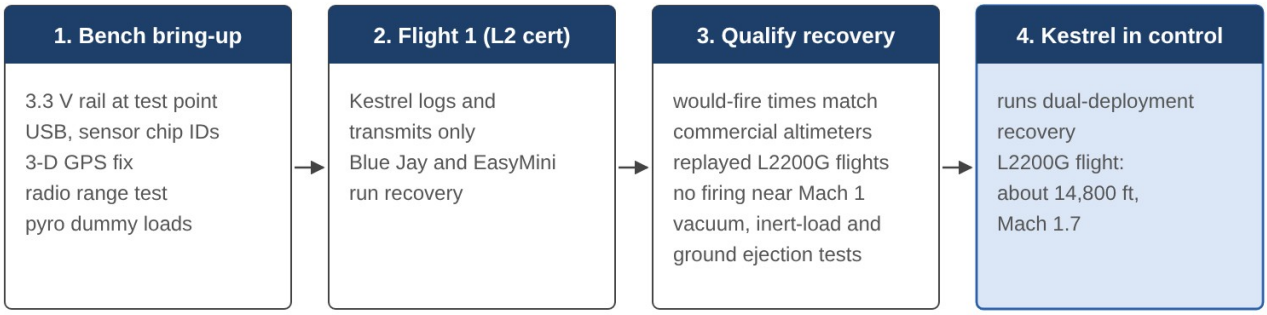
Pull-down resistors hold every gate off while the processor boots, and a small sense current through each e-match shows continuity without firing it.



One of the four pyro channels

Because my budget allows one board revision, I checked each subsystem against a datasheet or reference design before ordering, and planned a specific test to prove it once the boards arrive.

Subsystem	Checked against	Proven at bring-up by
Power	TI's TPS54202 3.3 V reference values; USB-only start-up margin	3.25–3.35 V at the test point
Processor	All 80 RP2350B pins against the pin table and SparkFun's production board	Enumerates over USB
Sensors and GPS	Each pinout and SPI mode against its datasheet; u-blox integration manual	Chip ID self-test; 3-D GPS fix
Radio	Pinout; link budget of about 29 dB margin at 10 km	Ground-station range test
Pyro	MOSFET ratings, gate drive, boot pull-downs	Dummy-load tests, armed and disarmed
Layout	Zero ERC/DRC errors; outside review; PCBWay engineering review	Visual inspection of both assembled boards

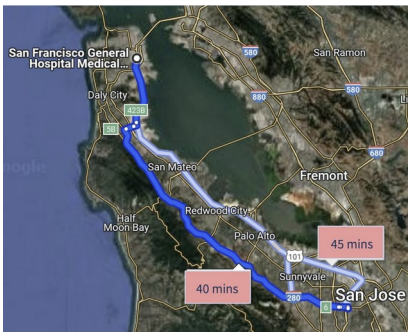


Test plan before Kestrel controls recovery

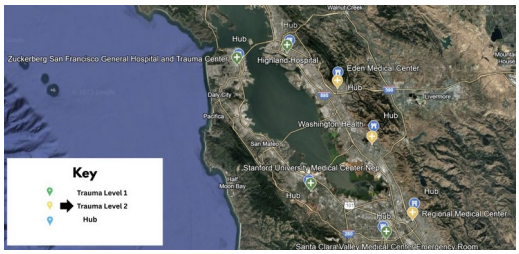
On the first flight Kestrel only logs and transmits, while a Featherweight Blue Jay and an EasyMini handle recovery. Before it controls a parachute, its would-fire times from flight 1 have to match the commercial altimeters, it has to fire correctly on replayed simulated L2200G flights and never during the Mach transition, and it has to pass vacuum-chamber, inert-load and ground ejection tests.

AEROPULSE - NASA AMES RESEARCH CENTER

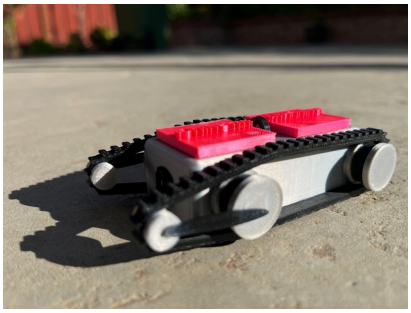
Concept Design Intern
June – August 2025



Modeled route (map data © Google)



Proposed drone hubs (imagery © Google)



Sub-scale robot prototype

What?

- Our team of six designed **Aeropulse**, a concept that moves organs, lab samples and urgent medicine between hospitals with hybrid VTOL drones and stair-climbing robots, with no manual handoffs
- I was one of 36 students picked from about 800 applicants for the Aviation Systems Division's Concept Design Experience

How?

- Examined routing constraints, urban airspace integration (UTM, geofencing, detect-and-avoid) and trade-offs between the drones and the ground robots
- Conceptualized route planning for both vehicles, with 3-D occupancy-grid routing, GPS and SLAM for the drones and A* and Dijkstra path planning for the robots
- Planned for operations below 400 ft AGL under **Part 107** with BVLOS waivers

Results

- In our model, a donor heart gets from San Francisco General to Santa Clara Valley Medical Center in about 37 min, compared with 70 to 90+ min today
- Presented the design to 50+ NASA staff
- Co-authored NASA/TM-20250009029 with the rest of the team

THE CONCEPT

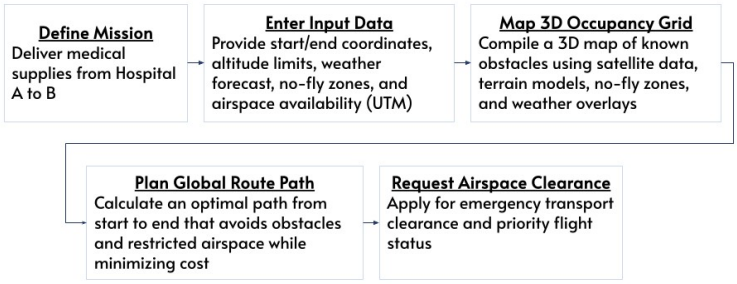


A robot carries the payload out of Hospital A, rides the drone, and takes it to the right room in Hospital B (from the memo)

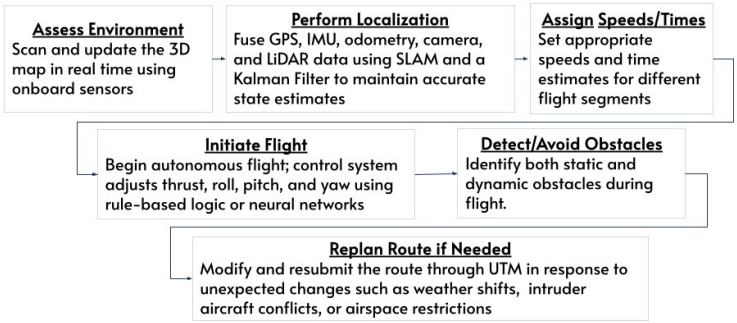
ROUTE PLANNING (MY FOCUS AREA)

Before each mission the drone plans its route from Hospital A to Hospital B. The inputs are the start and end coordinates, altitude limits, the weather forecast, no-fly zones and UTM airspace availability.

These are compiled into a 3-D occupancy grid, the global path is planned around obstacles and restricted airspace at the lowest cost, and the drone requests emergency priority clearance through UTM.



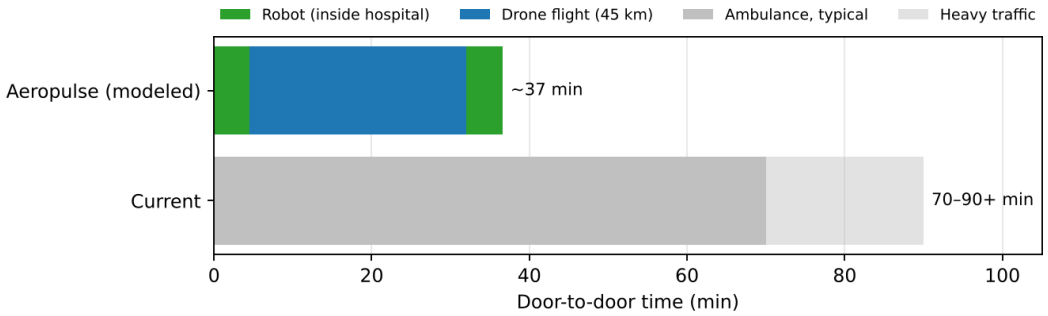
Pre-flight route planning (from the memo)



In-flight route planning (from the memo)

In flight, the drone keeps its map updated and fuses GPS, IMU, odometry, camera and LiDAR data with SLAM and a Kalman filter to track where it is.

If the weather shifts, another aircraft conflicts or a new restriction appears, it replans and resubmits the route through UTM. If it can't continue safely, it can loiter, land at a low-risk site, return to base or deploy a parachute.

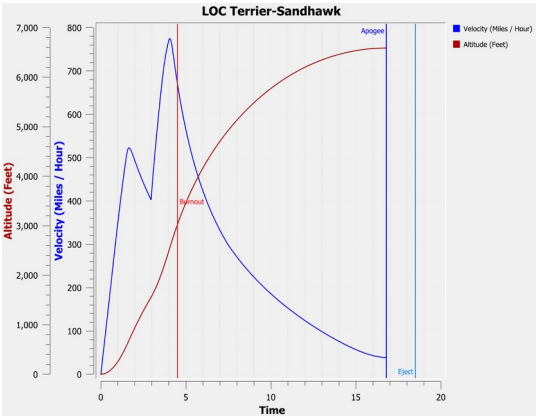


Modeled heart transfer, San Francisco General to Santa Clara Valley Medical Center

Full memo: [ntrs.nasa.gov/citations/20250009029 (NASA/TM-20250009029)](ntrs.nasa.gov/citations/20250009029)

TERRIER-SANDHAWK TWO-STAGE ROCKET

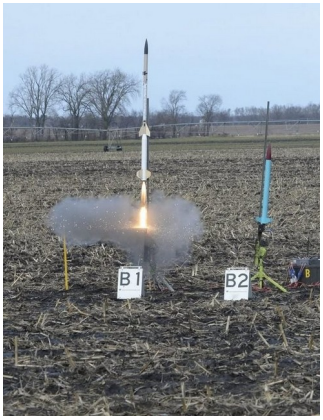
Builder and Flyer, Personal Project
January 2025



RockSim simulation



Setting up on pad B1



Liftoff

What?

- My first two-stage high-power rocket, a LOC Precision 2.26" Sandhawk sustainer on a Terrier booster (76.9 in, 2,078 g)
- Flew it on a Cesaroni I297 and H180 at Michiana Rocketry on January 31, 2025

How?

- Built the avionics for electronic staging, sustainer air-start, GPS tracking and telemetry. The stages separate 1.0 s after booster burnout and the sustainer lights 0.7 s later
- Simulated the flight in **RockSim** to choose motors and check stability (2.40 static margin)
- Analyzed how the 0.7 s ignition delay, which RockSim doesn't model, changes the trajectory using a 2-D point-mass model

Results

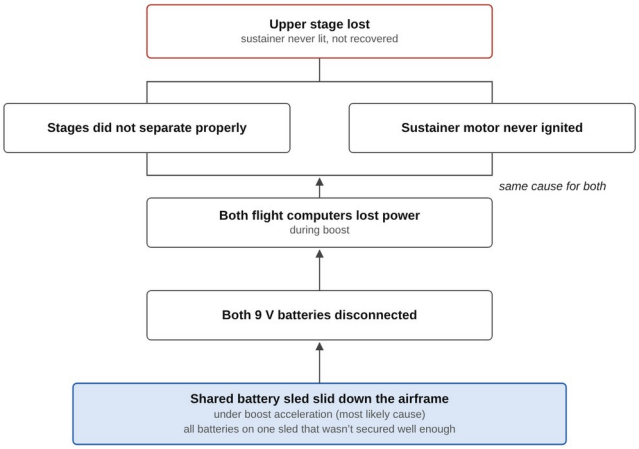
- The delay raised predicted apogee from about 6,590 ft to 6,860 ft
- On the flight the stages didn't separate properly, the sustainer never lit and the upper stage was lost
- Both 9 V batteries were on one sled that most likely slid under boost and disconnected them, which taught me the importance of redundant avionics

FAILURE ANALYSIS

The stages didn't separate properly and the sustainer never lit. Since both happened on the same flight, I looked for something that could cause both.

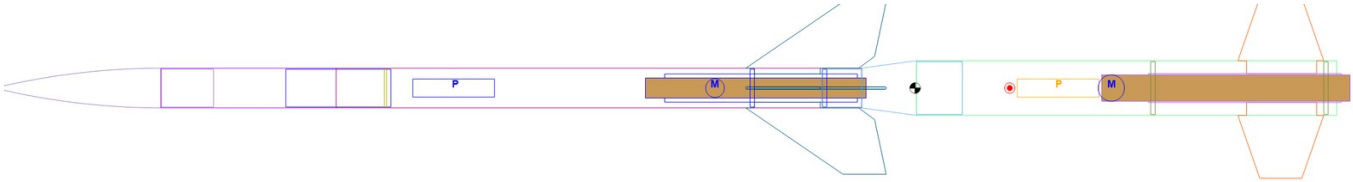
Every flight computer ran on a 9 V battery, and all of the batteries were mounted to the same sled, which wasn't secured well enough. Under boost acceleration the sled most likely slid down the airframe and disconnected both batteries, so all of the electronics lost power at once.

This flight taught me the importance of redundant avionics systems. Having more than one flight computer didn't help, because they all depended on the same mount.



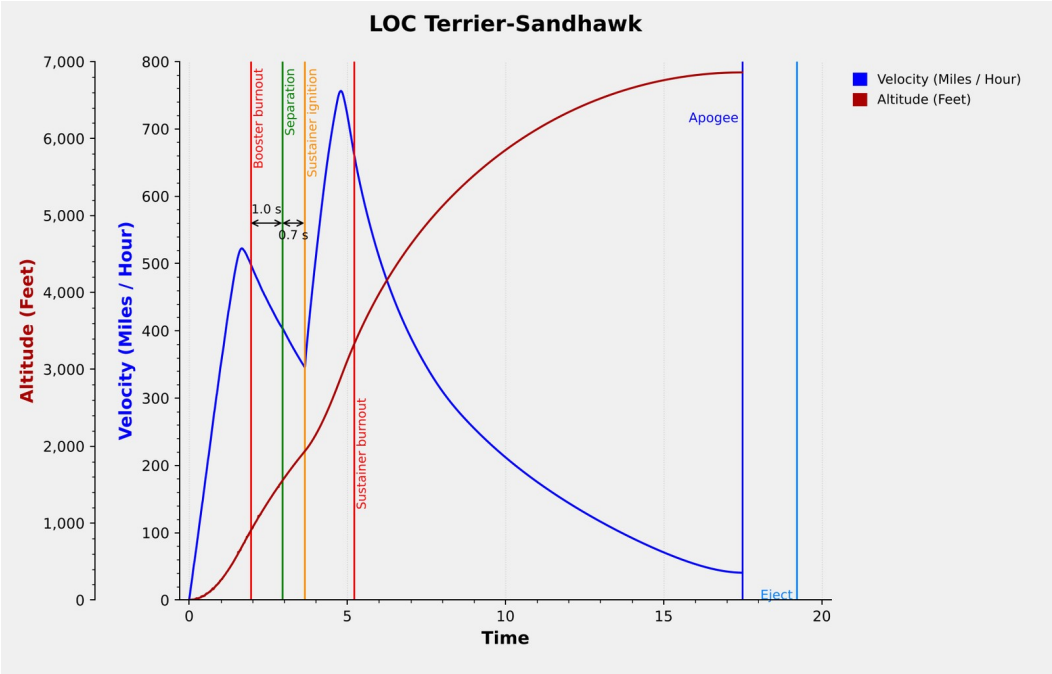
How one loose sled led to both failures

STAGING DELAY ANALYSIS

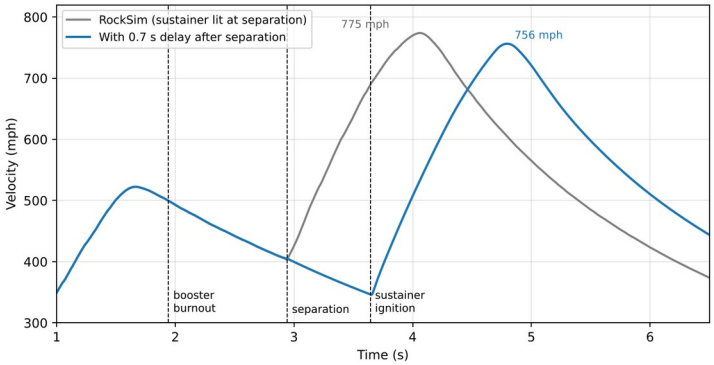


RockSim model, sustainer on the left (CG 52.5 in, CP 57.9 in from the nose)

RockSim lights the sustainer at the moment the stages separate, so its prediction does not include the 0.7 s wait before sustainer ignition. To model the actual sequence, the upper stage was re-flown from separation with a two-dimensional point-mass model. The sustainer's drag-to-mass ratio was taken from RockSim's coast phase as a function of Mach number, and its ignition mass was fitted to the boost phase using the H180's certified thrust curve. Run with RockSim's own timing, the model matches RockSim's apogee within 3 ft and its peak speed within 0.3%.



Predicted flight with the 1.0 s separation and 0.7 s ignition delays



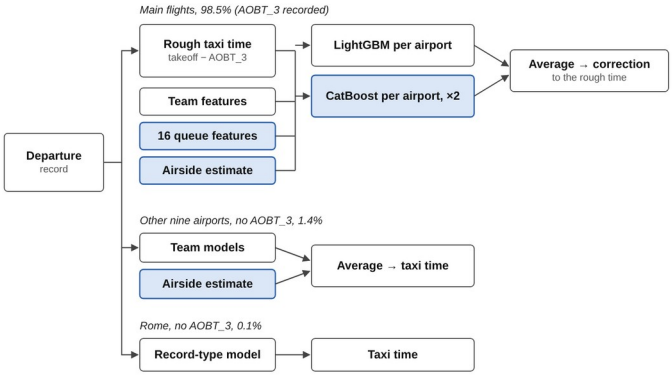
RockSim (gray) vs. delayed ignition (blue)

With the delay, the sustainer lights at 347 mph instead of 415 mph and peaks at 756 mph instead of 775 mph. Less of the burn happens near Mach 1, where drag is highest, so the predicted apogee rises from about 6,590 ft to 6,860 ft.

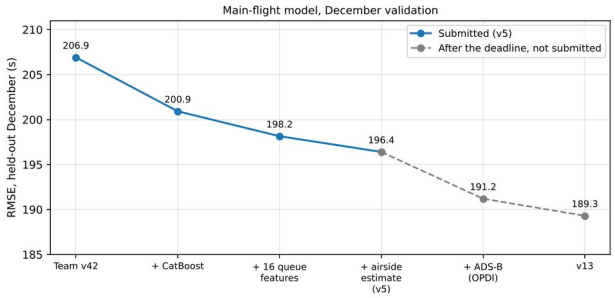
A 10% change in drag moves this by about 300 ft. Because the sustainer never lit on this flight, the prediction couldn't be checked against flight data.

EUROCONTROL PRC DATA CHALLENGE

Team Member, Embry-Riddle Team
September – October 2026



Team model, my additions in blue



Held-out error by version (gray points came after the deadline)

What?

- Predict aircraft taxi-out time at ten major European airports from about 2 million 2025 departures, scored by RMSE on January and July 2026 flights
- I joined Embry-Riddle's team "merry-llama" with very little Python or machine learning experience and worked on the model for main flights (98.5% of departures)

How?

- Added **CatBoost** models alongside the team's LightGBM models
- Built 16 departure-queue features, like how many planes took off from the same runway in the previous 5 to 20 minutes
- Used **SHAP** on the 1,000 worst predictions, then trained an "airside" taxi-time estimate that doesn't rely on the reported off-block time
- Decided what counted as an improvement before each test, and kept notes on what worked and what didn't across 20+ experiments

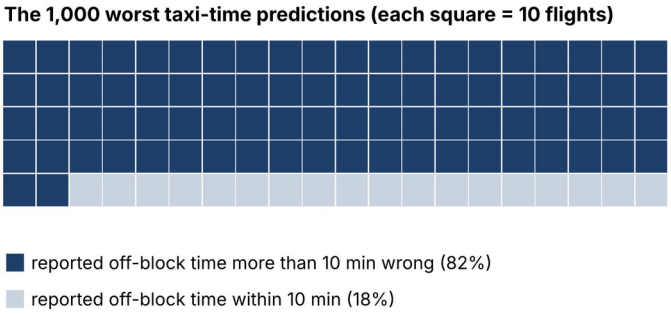
Results

- Lowered the team's official RMSE from 276.53 s to 267.34 s
- Taeyun Yoo built the team's best score (265.02 s) and the final pipeline on top of this version, and the final pipeline still uses my CatBoost models and airside estimate

APPROACH

I started from the team's LightGBM model for main flights and tested one addition at a time against held-out data. Before each test I decided what result would count as an improvement.

Looking at the worst predictions with SHAP, most came from flights whose reported off-block time was more than 10 minutes off. That's why I trained an airside estimate that predicts taxi time without it.



Worst 1,000 predictions

Team code: github.com/taeyunyoo/prc-2026-merry-llama